

Explainable Artificial Intelligence for Cancer Detection Using Medical Imaging and Clinical Data

Sarfo K. Frimpong
Master's of Computer Science

Department of Computer Science, Virginia State University
Petersburg, Virginia, USA
sfri0013@students.vsu.edu

Abstract— This study presents an explainable deep learning framework for automated malignant detection in breast ultrasound imaging. Using the Breast Ultrasound Dataset (BUSI) [11], a transfer learning-based ResNet-18 model [13] was fine-tuned for binary classification of malignant versus non-malignant cases. On a held-out test set ($n = 117$), the model achieved 94.87% accuracy, 87.10% sensitivity, 97.7% specificity, and an area under the ROC curve (AUC) of 0.9816, indicating strong class separability. To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) [14] was applied and qualitatively compared with expert-annotated lesion masks. Results demonstrate strong alignment between model attention and pathological regions. This work emphasizes sensitivity-driven evaluation and explainability integration, advancing clinically responsible AI-based decision-support systems for breast cancer detection.

Keywords—Breast Cancer, Ultrasound Imaging, Deep Learning, Convolutional Neural Networks, ResNet-18, Transfer Learning, Explainable Artificial Intelligence, Grad-CAM, Medical Image Classification, Clinical Decision Support

CHAPTER 1

I. INTRODUCTION

Cancer remains one of the leading causes of mortality worldwide, with breast cancer representing one of the most prevalent malignancies affecting women [1], [2]. Early detection is the most critical factor influencing survival outcomes. When identified at early stages, five-year survival rates exceed 90%, whereas late-stage diagnosis significantly reduces prognosis and increases treatment complexity [2], [3].

Breast ultrasound imaging plays an essential role in diagnostic workflows, particularly for patients with dense breast tissue where mammography sensitivity may be limited [4], [5]. Ultrasound is non-ionizing, cost-effective, and widely accessible, making it an important complementary modality in breast cancer screening and diagnostic evaluation [5]. However, interpretation remains highly dependent on radiologist expertise and is subject to inter-observer variability caused by operator skill, image quality, and diagnostic fatigue [6]. As screening volumes increase globally, healthcare systems face growing demand for consistent and scalable diagnostic support mechanisms, motivating the integration of computer-aided diagnostic systems [6], [7].

These challenges motivate the development of computational systems capable of augmenting radiological assessment, improving diagnostic reliability, and reducing human variability in interpretation.

II. ARTIFICIAL INTELLIGENCE IN BREAST CANCER DETECTION

Artificial intelligence (AI), particularly deep learning, has demonstrated significant promise in medical image analysis [7]. Convolutional neural networks (CNNs) automatically learn hierarchical feature representations from raw pixel data, eliminating the need for handcrafted feature engineering [7].

In oncology, CNN-based models have achieved strong performance across multiple imaging modalities, including mammography, computed tomography (CT), dermoscopy, and ultrasound imaging [8], [10]. Large-scale studies have shown that deep learning systems can approach or exceed specialist-level performance in selected diagnostic tasks [8], [10]. Due to limited dataset sizes in medical imaging, transfer learning using pretrained architectures such as ResNet has become a standard approach to improve convergence stability and generalization performance [7].

However, predictive accuracy alone is insufficient for clinical deployment. AI systems must demonstrate reliability, transparency, and clinically meaningful evaluation to ensure safe and responsible integration into healthcare workflows [6], [9].

III. PROBLEM DEFINITION

This study investigates automated detection of malignant lesions in breast ultrasound images using the Breast Ultrasound Dataset (BUSI), which contains 780 images: 437 benign, 210 malignant, and 133 normal. [11]

A. Data Preprocessing

All images were resized to 224×224 pixels to match the ResNet-18 input requirements and normalized to the range $[0,1]$ to stabilize training. Data augmentation, including random horizontal flipping, rotation, and scaling, was applied to the training set to improve generalization. Validation and test sets were not augmented to ensure unbiased evaluation. Although contrast enhancement methods such as histogram equalization and CLAHE were explored, the final model used only normalization and geometric augmentation to preserve clinical image characteristics.

ResNet-18 was selected due to its balance between model complexity and computational efficiency, making it well suited for medical imaging tasks with limited datasets. Its residual learning framework enables effective training of deeper representations while mitigating vanishing gradient issues, leading to stable convergence and improved feature extraction [13]. Additionally, the availability of pretrained weights allows transfer learning, which enhances performance and generalization in small-scale medical datasets such as BUSI [7], [13].

B. Problem Formulation

$$\text{Malignant}=1, \text{Benign} + \text{Normal}=0$$

After binary conversion this resulted in:

- 570 negative samples
- 210 positive samples

C. Dataset Split

The dataset was split into training, validation, and test subsets. The held-out test set contained 117 images:

- 86 negative
- 31 malignant

D. Model Performance

On the test set, the fine-tuned ResNet-18 model achieved an overall accuracy of 94.87%, with precision of 93.10%, sensitivity (recall) of 87.10%, **specificity of 97.7%**, an F1-score of 0.9000, and a receiver operating characteristic area under the curve (ROC-AUC) of 0.9816. The corresponding confusion matrix revealed 84 true negatives, 27 true positives, 2 false positives, and 4 false negatives. [12]:

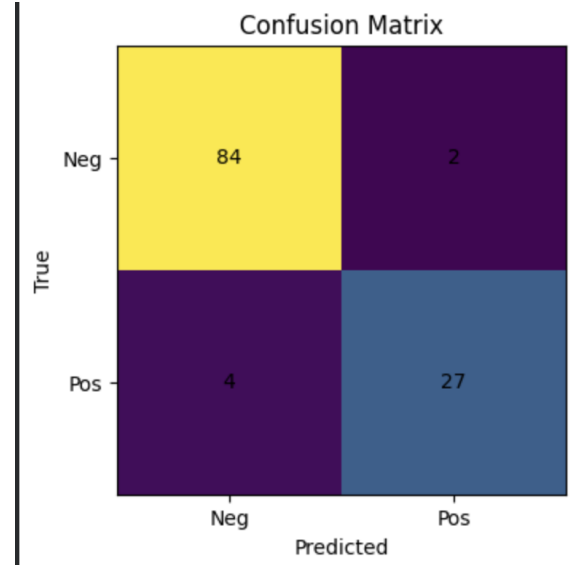


Figure 1.0 Confusion Matrix

IV. EXPLAINABILITY AND CLINICAL ALIGNMENT

This study integrates explainability to align deep learning predictions with clinical reasoning. Grad-CAM is used to generate class-discriminative localization maps highlighting regions that influence model predictions [14].

These activation maps are qualitatively compared with expert-annotated lesion masks from the BUSI dataset to evaluate spatial correspondence with ground-truth pathological regions [11]. This lesion-level comparison provides a proxy for assessing clinical plausibility.

Analysis of misclassified cases reveals limitations in scenarios involving low-contrast lesions, indistinct boundaries, and speckle noise inherent to ultrasound imaging. Grad-CAM

supports error analysis by exposing model attention patterns, improving transparency and trust in AI-assisted diagnosis.

V. RESEARCH AIM AND CONTRIBUTIONS

The objective of this thesis is to develop an explainable deep learning framework for breast ultrasound cancer detection that balances predictive performance with interpretability. Although deep learning models achieve strong performance in medical imaging [7], [8], [10], their lack of transparency limits clinical adoption [6], [9].

A standardized preprocessing pipeline was applied, including image resizing (224×224), normalization, and controlled augmentation to improve generalization under limited data conditions.

This study makes the following contributions:

- i. **Transfer Learning Framework:**
A fine-tuned ResNet-18 model is used for binary classification, leveraging pretrained features to improve performance and optimization stability [13].
- ii. **Specificity-Driven Evaluation:**
Performance is evaluated using clinically relevant metrics, with emphasis on specificity to reduce false positives and accurately identify non-malignant cases, while also reporting sensitivity (recall), accuracy, precision, and ROC-AUC.
- iii. **Explainability Integration:**
Grad-CAM is applied to generate class-specific activation maps, providing insight into model predictions [14].
- iv. **Lesion-Level Validation:**
Activation maps are compared with expert-annotated lesion masks to assess spatial alignment with clinically relevant regions [11].

This work emphasizes reproducibility, robustness, and clinical alignment. Although the model demonstrates high specificity in reducing false positives and correctly identifying true negatives, sensitivity is also evaluated to mitigate the risk of missed malignant cases, ensuring a clinically meaningful assessment.

CHAPTER 2 - ANALYTICAL AND REVIEW FOUNDATION

VI. DEEP LEARNING IN BREAST ULTRASOUND CLASSIFICATION

Deep learning has significantly transformed medical image analysis. In breast cancer diagnostics, convolutional neural

networks (CNNs) have demonstrated strong performance across mammography, ultrasound, and MRI modalities. Unlike traditional machine learning systems that rely on handcrafted features, CNNs learn hierarchical representations directly from image data through end-to-end optimization.

Breast ultrasound presents unique challenges for automated classification. Images are characterized by:

- Speckle noise
- Low contrast between lesions and surrounding tissue
- Irregular lesion boundaries
- Operator-dependent acquisition variability

These factors increase intra-class variability and reduce signal consistency, making ultrasound classification more complex than other modalities.

Although CNN modal report accuracy between 85% and 95%, results vary considerably across datasets due to differences in preprocessing, class imbalance handling, and evaluation protocols. Importantly, many studies emphasize overall accuracy without adequate focus on sensitivity. In malignant detection, false negatives carry significant clinical risk, making sensitivity a more meaningful metric than accuracy alone.

VII. RESIDUAL NETWORKS AND TRANSFER LEARNING

Residual Networks (ResNets) introduced shortcut connections that improve gradient flow and enable stable training of deeper architectures. The residual formulation:

$$H(x) = F(x) + x$$

In a Residual Network (ResNet), instead of learning a full transformation $H(x)$, the network learns a residual function $F(x)$, which represents the difference between the input and the desired output. The input x is then added directly to the learned residual [13].

In medical imaging, transfer learning is widely adopted due to limited dataset sizes. Pretraining on ImageNet enables models to learn generalizable low-level features that transfer effectively to ultrasound imaging.

ResNet-18 was selected in this study due to its balance between representational capacity and computational efficiency. Fine-tuning achieved:

- 98.17% training accuracy
- 92.31% validation accuracy

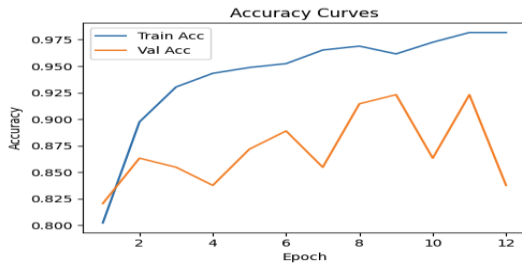


Figure 1.1 Training Accuracy Curve Graph

The modest generalization gap indicates controlled overfitting and effective adaptation to ultrasound-specific features.

VIII. CLINICAL EVALUATION AND RISK-ORIENTED METRICS

In diagnostic medicine, evaluation must reflect clinical risk priorities. Accuracy can obscure class imbalance effects, particularly when negative cases dominate.

Specificity:

- TN (True Negatives) = Non-malignant cases correctly identified as non-malignant
- FP (False Positives) = Non-malignant cases incorrectly classified as malignant

$$\frac{TN}{TN + FP}$$

is critical because false positives can lead to unnecessary follow-up procedures, patient anxiety, and increased healthcare costs.

Sensitivity:

- TP (True Positives) = Malignant cases correctly identified as malignant
- FN (False Negatives) = Malignant cases incorrectly classified as benign

$$\frac{TP}{TP + FN}$$

is critical because false negatives represent missed cancer diagnoses. Example:

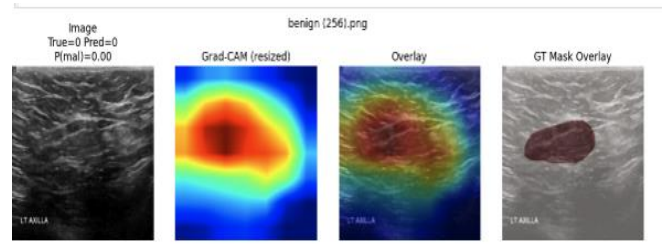


Figure 1.2 Missed cancer diagnoses of malignancies and benign conditions. True=1, Pred=0 P(mal)=0.00

Initial experiments achieved:

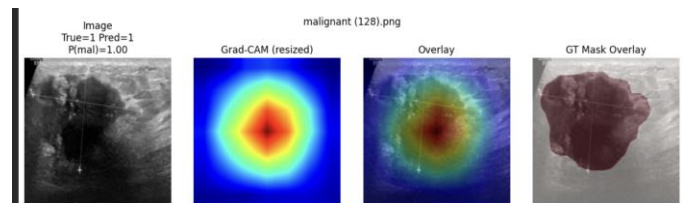
- Sensitivity: 80.6%
- Specificity: ~94%
- ROC-AUC: 0.9546

The confusion matrix revealed six false negatives, highlighting improvement needs.

After optimization, performance improved to:

- Accuracy: 94.87%
- Sensitivity: 87.10%
- Specificity: 97.7%
- ROC-AUC: 0.9816

False negatives decreased to four, while specificity increased, demonstrating improved discrimination between malignant and non-malignant cases. The high specificity indicates that the model effectively minimizes false positives and reliably identifies benign conditions, which is essential for reducing unnecessary clinical interventions [9].



Although sensitivity remains slightly lower than specificity, it is maintained at a clinically meaningful level to reduce missed diagnoses. Reporting class-wise metrics and confusion matrices aligns evaluation with clinical diagnostic priorities rather than relying solely on aggregate accuracy [6], [7].

Figure 1.3 True Positive diagnoses of malignant cancer. True=1 Pred=1 P(mal)=1.00

IX. EXPLAINABLE ARTIFICIAL INTELLIGENCE IN ONCOLOGY

Deep learning models are often criticized for limited interpretability. In oncology, transparency is essential because diagnostic decisions directly affect treatment planning.

Grad-CAM generates class-discriminative localization maps:

$$L_{Grad-CAM}^c = ReLu\left(\sum_K a_k^c A^k\right)$$

where feature maps are weighted by class-specific gradients.

In this study, Grad-CAM was applied to the final convolutional layer of ResNet-18 and overlaid on ultrasound images to visualize discriminative regions.

X. LESION-LEVEL VALIDATION AND RESEARCH POSITIONING

A key advantage of the BUSI dataset is the availability of expert-annotated lesion masks [11]. Many prior studies incorporate visualization techniques without validating whether highlighted regions correspond to true pathological structures [7], [14].

By comparing Grad-CAM heatmaps with ground-truth masks, this study performs lesion-level validation. True negative and true positive cases demonstrate localized activation over relevant regions, while misclassified cases often exhibit partial, diffuse, or spatially shifted attention.

This dual evaluation strategy, combining quantitative classification metrics with spatial localization analysis, addresses three key gaps in existing literature:

- Overemphasis on accuracy without **specificity-focused analysis** [6], [9]
- Limited integration of explainability in model evaluation [14], [15]
- Lack of lesion-level attention validation [7]

By integrating transfer learning, **specificity-driven evaluation**, Grad-CAM visualization, and mask-based validation, this work advances toward clinically meaningful and interpretable AI systems for breast cancer detection [13], [14].

Although the model emphasizes specificity to reduce false positives and improve reliable identification of non-malignant cases, sensitivity is also considered to mitigate the clinical risk of missed malignant diagnoses, ensuring a balanced and clinically responsible evaluation [6], [9].

CHAPTER 3 – RESEARCH FRAMEWORK AND EXPERIMENTAL DESIGN

a. Research Problems

Breast ultrasound imaging plays a critical role in breast cancer detection, particularly for patients with dense breast tissue where mammography may exhibit reduced sensitivity. Despite its clinical importance, ultrasound interpretation remains highly dependent on radiologist expertise and is affected by image noise, lesion variability, and operator-dependent acquisition. These factors introduce diagnostic variability and may lead to missed malignancies or unnecessary follow-up procedures [6].

Recent advances in artificial intelligence, particularly convolutional neural networks (CNNs), have demonstrated strong capability in medical image classification tasks [7]. However, several challenges remain before such systems can be reliably integrated into clinical workflows. First, many AI-based diagnostic models prioritize overall accuracy while neglecting clinically critical metrics such as sensitivity or . In cancer detection, false negatives correspond to missed malignant lesions and therefore represent the most serious diagnostic risk.

Second, deep learning systems often lack interpretability. Clinical adoption requires transparency to ensure that model predictions are based on meaningful pathological features rather than spurious image artifacts [9]. Without visual explanation mechanisms, clinicians cannot easily verify whether automated predictions align with anatomical structures associated with disease.

This research therefore investigates whether an explainable deep learning model can achieve high diagnostic performance while maintaining interpretability and clinical plausibility in breast ultrasound cancer detection.

b. Dataset and Experimental Setting.

The experiments were conducted using the **Breast Ultrasound Dataset (BUSI)** [11], which contains ultrasound images categorized into benign, malignant, and normal classes. To align the task with clinically meaningful malignant detection, the dataset was reformulated as a **binary classification problem**, where malignant cases represent positive samples and benign or normal cases represent negative samples.

After preprocessing and dataset preparation, the final dataset distribution consisted of:

- **570 negative samples** (benign + normal)
- **210 malignant samples**

The dataset was divided into **training, validation, and testing subsets**:

Dataset split	Dataset Distribution		
	Total Images	Negative	Positive
Training	546	399	147
Validation	117	85	32
Test	117	85	31

The held-out **test set of 117 images** was used exclusively for final performance evaluation

c. Model Architecture and Training

A **transfer learning approach using ResNet-18** was employed. ResNet uses residual connections that allow deeper networks to train effectively by learning residual mappings:

$$H(x) = F(x) + x$$

The model was initialized with **ImageNet pretrained weights**, enabling transfer of low-level visual features such as edges and textures to the medical imaging task.

Training was conducted for **12 epochs**. Training accuracy gradually increased to approximately **98%**, while validation accuracy stabilized around **92–93%**. Training loss decreased consistently across epochs, while validation loss fluctuated slightly but remained stable, suggesting reasonable generalization with limited overfitting.

d. Model Explainability

To improve interpretability, **Grad-CAM (Gradient-weighted Class Activation Mapping)** was applied to visualize the regions of the ultrasound images influencing model predictions.

Grad-CAM generates localization maps using gradients from the final convolutional layer:

$$L_C^{Grad-CAM} = ReLu \left(\sum_k a_k^c A^k \right)$$

The resulting heatmaps were overlaid on ultrasound images to highlight areas of model attention.

Visual analysis showed that for correctly classified malignant cases, the Grad-CAM heatmaps focused on lesion regions that correspond closely with expert-annotated segmentation masks. This suggests that the model learned clinically meaningful features associated with tumor morphology.

For benign cases, activation regions were typically diffuse or low-intensity, reflecting the absence of malignant patterns. Misclassified samples occasionally showed attention outside the lesion region, indicating potential sources of prediction error.

e. Research Questions

This study addresses the following research questions:

RQ1:

Can a transfer learning-based ResNet-18 model accurately classify malignant breast ultrasound images?

RQ2:

Does sensitivity provide a clinically meaningful evaluation of model performance?

RQ3:

Do Grad-CAM visualizations correspond to lesion regions identified by expert annotations?

f. Research Hypothesis

The following hypotheses guide this research:

H1:

A transfer learning-based ResNet-18 model can achieve diagnostic accuracy above 90%.

H2:

Model optimization improves sensitivity in detecting malignant lesions.

H3:

Grad-CAM heatmaps align with expert-annotated lesion regions in true positive predictions

CHAPTER 4 – RESULT AND DISCUSSION

a. Research Problems

The ResNet-18 transfer learning model was trained for 12 epochs using the BUSI breast ultrasound dataset [11]. Training accuracy increased from approximately 80% to 98%, while validation accuracy stabilized between 84% and 92%, indicating effective feature learning and reasonable generalization. Training loss decreased consistently, whereas validation loss showed moderate fluctuations, which is common in medical imaging tasks with limited and imbalanced datasets [7].

Beyond these trends, the gap between training and validation performance suggests mild overfitting, likely due to dataset size and variability in ultrasound image quality. However, the absence of significant divergence indicates that transfer learning with pretrained architectures such as ResNet-18 [13] effectively stabilizes optimization and improves generalization.

b. Test Set Evaluation

The final model was evaluated using a held-out **test dataset consisting of 117 ultrasound images**, ensuring unbiased performance assessment.

The model achieved the following classification performance:

Metric	Value
Accuracy	94.87%
Precision	93.10%
Sensitivity	87.10%
F1 Score	0.90
ROC-AUC	0.9816

The high ROC-AUC indicates excellent separability between malignant and non-malignant classes. However, performance analysis reveals an important clinical trade-off: while overall accuracy and precision are high, sensitivity is comparatively lower, indicating that some malignant cases were missed.

This imbalance suggests that the model is slightly biased toward predicting the majority (non-malignant) class, which is a common challenge in medical datasets with class imbalance.

c. Confusion Metrics

The model achieved high specificity (~97.7%), with only two false positives. However, four malignant cases were misclassified as benign (false negatives), reducing sensitivity to 87.1%:

	Predicted Negatives	Predicted Postives
True Negatives	84	2
True Positives	27	4

i. Why False Negatives Occur.

False negatives are particularly critical in medical diagnosis. In this study, they may be attributed to:

- **Low contrast lesions:** Some malignant regions have intensity patterns like surrounding tissue, making them difficult to distinguish.
- **Small lesion size:** Small or early-stage tumors may not provide sufficient spatial features for the model to detect.
- **Irregular or diffuse boundaries:** Ultrasound images often contain speckle noise and unclear margins, which can obscure lesion structure.

1) Difficult Sample Characteristics

The most challenging samples typically include:

- Images with **poor image quality or noise artifacts**
- Lesions with **weak echogenic differences**
- Cases where **benign and malignant features overlap visually**

These factors highlight limitations in relying solely on pixel-based learning without additional clinical context

d. Imaging Limitations and Clinical Implications

Ultrasound imaging inherently presents challenges such as speckle noise, operator dependency, and low contrast resolution compared to modalities like CT or MRI. These limitations directly affect model performance and contribute to misclassification errors.

From a clinical perspective, minimizing false negatives is critical, as missed malignant cases can delay diagnosis and treatment. While the model demonstrates strong overall performance, improving sensitivity remains a key area for future work.

e. Explainability and Clinical Alignment

Grad-CAM visualization shows that, in correctly classified cases, the model focuses on regions corresponding to annotated lesions. This alignment between model attention and expert-defined pathology supports the reliability of the model’s predictions.

However, in false negative cases, Grad-CAM often highlights incomplete or incorrect regions, suggesting that the model fails to fully capture subtle lesion features. This reinforces the importance of explainability in identifying model weaknesses and guiding improvements.

B. Summary of Findings

The ResNet-18 transfer learning model achieved 94.87% accuracy and a ROC-AUC of 0.9816, demonstrating strong classification capability. The model exhibited high specificity

with low false positive rates, while sensitivity was slightly lower due to missed malignant cases.

Overall, these results indicate that explainable deep learning models can provide accurate and interpretable diagnostic support. However, challenges such as false negatives, difficult sample characteristics, and inherent ultrasound limitations highlight the need for continued refinement toward clinically robust AI systems.

CHAPTER 5 – CONCLUSION AND FUTURE WORK

i. Conclusion

Breast cancer remains one of the most common cancers worldwide, and early detection is critical for improving survival outcomes. Ultrasound imaging is widely used for screening, particularly in patients with dense breast tissue, but interpretation can be challenging due to image noise, lesion variability, and operator dependency.

This study presents an explainable deep learning approach for breast ultrasound cancer detection using a ResNet-18 transfer learning model trained on the BUSI dataset. The task was formulated as a binary classification problem to distinguish malignant from non-malignant lesions.

The model achieved strong performance on the test set, with 94.87% accuracy, 93.10% precision, 87.10% recall, and a ROC-AUC of 0.9816, demonstrating effective discrimination between classes.

To improve interpretability, Grad-CAM was applied to visualize model attention. The resulting heatmaps showed strong alignment with annotated lesion regions, indicating that predictions were based on clinically relevant features.

Overall, these results demonstrate that explainable convolutional neural networks can provide accurate and interpretable support for breast cancer detection, with potential to improve diagnostic consistency and reduce missed malignancies.

ii. Research Conclusions

This research contributes to the application of artificial intelligence in medical imaging in several ways.

First, a deep learning framework for breast ultrasound classification was developed using a transfer learning approach with ResNet-18, enabling effective feature learning in a limited-data setting.

Second, the model achieved strong diagnostic performance on the BUSI dataset, with high accuracy and ROC-AUC, demonstrating reliable malignant detection.

Third, Grad-CAM was integrated to provide visual explanations of model predictions. Comparison of attention maps with lesion masks showed alignment with clinically relevant regions, supporting interpretability.

These contributions demonstrate the value of combining deep learning and explainable AI to improve transparency and trust in medical imaging systems.

C. Limitations

Despite promising results, several limitations remain. The dataset size is relatively small, which may limit generalization to larger and more diverse clinical populations [7]. Additionally, images were collected under controlled conditions, and real-world variability may affect performance.

Finally, Grad-CAM provides qualitative explanations only. Quantitative evaluation of explainability remains an open challenge in medical AI.

D. Future Work

Future work can improve and extend this research in several directions. Evaluating the model on larger, multi-institutional datasets will improve robustness. Enhancing sensitivity through techniques such as data augmentation or cost-sensitive learning may reduce false negatives.

Multimodal approaches combining ultrasound with mammography or MRI could further improve diagnostic performance. Additionally, advanced explainability methods such as SHAP or attention-based models may provide deeper insights.

The framework can also be extended into clinical decision support systems to assist radiologists in real-time diagnosis [6].

E. Final Remarks

This study demonstrates that explainable convolutional neural networks can achieve high accuracy in breast ultrasound cancer detection while providing interpretable visual insights. The integration of deep learning and explainable AI offers a promising approach for improving diagnostic reliability and supporting clinical decision-making in medical imaging.

REFERENCES

- [1] World Health Organization (WHO), "Cancer fact sheet," 2023.
Available: <https://www.who.int/news-room/fact-sheets/detail/cancer>
- [2] American Cancer Society, *Breast Cancer Facts & Figures 2023–2024*, Atlanta, GA, USA, 2023.
Available: <https://www.cancer.org/research/cancer-facts-statistics/all-cancer-facts-figures/2023-cancer-facts-figures.html>
- [3] National Cancer Institute (NCI), "Cancer stat facts: Female breast cancer," SEER Program, 2023.
Available: <https://seer.cancer.gov/statfacts/html/breast.html>
- [4] P. Skaane, "Breast cancer screening with digital breast tomosynthesis," *Breast Cancer Research*, vol. 19, no. 1, pp. 1–9, 2017.
Available: <https://doi.org/10.1186/s13058-017-0859-3>
- [5] D. B. Kopans, *Breast Imaging*, 3rd ed. Philadelphia, PA, USA: Lippincott Williams & Wilkins, 2007.
Available: <https://books.google.com/books?id=breast-imaging>
- [6] K. Doi, "Computer-aided diagnosis in medical imaging: Historical review, current status, and future potential," *Computerized Medical Imaging and Graphics*, vol. 31, no. 4–5, pp. 198–211, 2007.
Available: <https://doi.org/10.1016/j.compmedimag.2007.02.002>
- [7] G. Litjens *et al.*, "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.
Available: <https://doi.org/10.1016/j.media.2017.07.005>
- [8] A. Esteva *et al.*, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.
Available: <https://doi.org/10.1038/nature21056>
- [9] R. B. Parikh *et al.*, "Addressing bias in artificial intelligence in health care," *NEJM Catalyst*, 2019.
Available: <https://doi.org/10.1056/CAT.19.1091>
- [10] P. Rajpurkar *et al.*, "Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practicing radiologists," *Nature Medicine*, vol. 24, no. 7, pp. 861–870, 2018.
Available: <https://doi.org/10.1038/s41591-018-0177-9>
- [11] A. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, "Dataset of breast ultrasound images," *Data in Brief*, vol. 28, Art. no. 104863, 2020.
Available: <https://doi.org/10.1016/j.dib.2019.104863>
Dataset: <https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset>
- [12] S. Frimpong, "BUSI explainable CNN with Grad-CAM," GitHub repository, 2026.
Available: https://github.com/frimpongsarfok/AI_Breast_Cancer_Classification_VSU/blob/main/BUSI_Explainable_CNN_GradCAM_Colab.ipynb
- [13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016.
Available: <https://doi.org/10.1109/CVPR.2016.90>
- [14] R. R. Selvaraju *et al.*, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE ICCV*, 2017.
Available: <https://doi.org/10.1109/ICCV.2017.74>
- [15] W. Samek, T. Wiegand, and K.-R. Müller, "Explainable artificial intelligence: Understanding, visualizing and interpreting deep learning models," *IEEE Signal Processing Magazine*, vol. 34, no. 6, pp. 56–71, 2017.
Available: <https://doi.org/10.1109/MSP.2017.2745539>